THE CHEMICAL RECORD

Catalytic Enantioselective Organozinc Addition Toward Optically Active Tertiary Alcohol Synthesis

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Received 14 February 2008; Accepted 6 March 2008

ABSTRACT: A highly enantioselective organozinc (R_2Zn) addition to a series of aldehydes and ketones was developed based on conjugate Lewis acid–Lewis base catalysis. Optically active secondary and tertiary alcohols were obtained in high yields with high enantioselectivities without Ti(IV) compounds. Bifunctional chiral 3,3′-diphosphoryl-BINOL ligands were designed and prepared through a phospho-Fries rearrangement as a key step. On the other hand, bifunctional chiral phosphoramide ligands were designed and prepared from L-valine. Mechanistic studies were performed by X-ray analyses of Zn(II) cluster and chiral ligands, a ^{31}P NMR experiment on Zn(II) complexes, and stoichiometric reactions with some chiral or achiral Zn(II) complexes to propose a transition state assembly that includes monomeric active intermediates. © 2008 The Japan Chemical Journal Forum and Wiley Periodicals, Inc. Chem Rec 8: 143–155; 2008: Published online in Wiley InterScience (www.interscience.wiley.com) DOI 10.1002/tcr.20146

Key words: alcohol; asymmetric catalysis; Lewis acid–Lewis base catalyst; organometallics; zinc(II)

Introduction

Optically active secondary and tertiary alcohols are versatile building blocks for the synthesis of natural products and pharmaceuticals. One important strategy for synthesizing these alcohols is a carbon–carbon bond-forming reaction between carbonyl compounds and organometallic reagents.[1,2] In particular, catalytic enantioselective organozinc addition to aldehydes is one of the most important reactions involving carbon–carbon bond formation, and optically active secondary alcohols are obtained.[3] However, it is still difficult to achieve a high yield and enantioselectivity for tertiary alcohols from ketones and organozinc reagents due to conspicuous steric and electronic constraints between ketones and organometallic reagents.[4] Furthermore, organometallic reagents often cause the enolization of ketones due to their high basicity or competitive reduction via β-H transfer to give undesired secondary alcohols. To date, numerous efficient chiral ligands have been developed for secondary alcohol synthesis from aldehydes, but their application to tertiary alcohol synthesis from ketones via alkylation or arylation is still extremely limited.[5–10] In principle, we can classify organometallic alkylation or arylation into three types: (i) double activation by only Lewis acidic organometallic reagents, (ii) double activation by a Lewis acid–Lewis base, and (iii) double activation by a conjugate Lewis acid–Lewis base (Scheme 1). Usually, organomagnesium and organoaluminum reagents belong to type (i). Most of the reported enantioselective organozinc additions to aldehydes are type (ii), where the Lewis acid and Lewis base are attached to each other.[11] In sharp contrast to types (i) and (ii), type (iii)

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catalysis involves an electron charge transfer at the ligand interior. Thus, type (iii) catalysis should offer a great advantage for enhancing catalytic activity because the direct linkage between the acid and the base in type (ii) may weaken or negate both activities. Recently, to overcome these essential problems in carbonyl compounds and organozinc reagents, we developed two type (iii) catalyses for the synthesis of optically active tertiary alcohols. One of these has a 3,3′-diphosphoryl-BINOL [BINOL = 1,1′-*bi*(2-naphthol)] skeleton and the other has a phosphoramide structure, which is derived from L-valine. Here, we report the development of a highly efficient enantioselective organozinc addition to not only aldehydes but also ketones using a conjugate Lewis acid–Lewis base catalytic system.[12,13]

Design of Chiral 3,3′-Diphosphoryl-BINOLate–Zn(II) Complexes[14]

In early investigations, it has been shown that the BINOLate–Ti(IV) catalyst is highly effective in diethylzinc addition to aldehydes, although more than a stoichiometric amount of $Ti(Oi\text{-}Pr)_4$ is necessary to achieve high yields and high enantioselectivities.[15] Moreover, 3,3′-disubstituted BINOLs have been designed for active Zn(II) catalysts in diethylzinc addition to aldehydes without $Ti(Oi\text{-}Pr)_4$.[16] However, sterically demanding 3,3′-disubstituted BINOLs are often difficult to synthesize in satisfactory yields because of the multistep transformations from the starting BINOL. To overcome this difficulty, we devised 3,3′-diphosphoryl-

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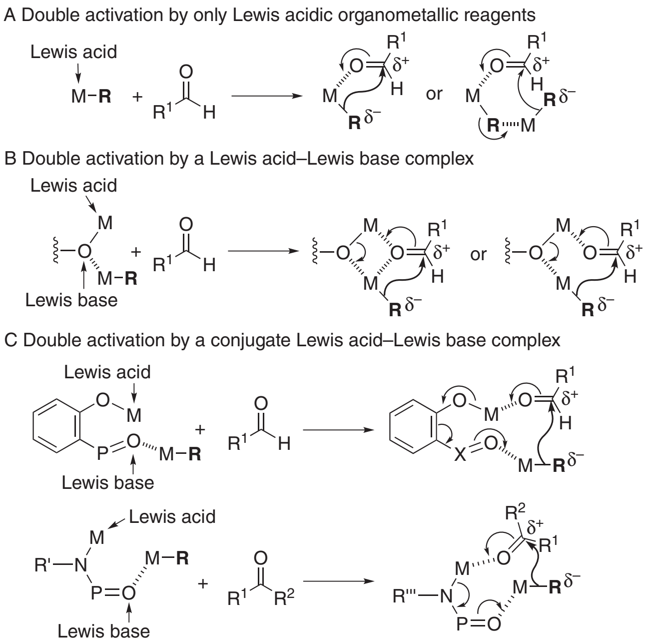
Kazuaki Ishihara was born in Aichi, Japan in 1963, and received his Ph.D. from Nagoya University in 1991 under the direction of Professor Hisashi Yamamoto. He had the opportunity to work under the direction of Professor Clayton H. Heathcock at the University of California, Berkeley, as a visiting graduate student for 3 months in 1988. He was a JSPS fellow under the Japanese Junior Scientists Program from 1989 to 1991. After completing his postdoctoral studies with Professor E. J. Corey at Harvard University (15 months beginning in 1991), he returned to Japan and joined Professor Hisashi Yamamoto's group at Nagoya University as an assistant professor in 1992, and became associate professor in 1997. In 2002, he was appointed to his current position as a full professor at Nagoya University. Dr. Ishihara received the Inoue Research Award for Young Scientists (1994); the Chemical Society of Japan Award for Young Chemists (1996); the Thieme Chemistry Journal Award (2001); the Green and Sustainable Chemistry Award from the Ministry of Education, Culture, Sports, Science and Technology (2003); the JSPS Prize (2005); the BCSJ Award (2005); the 0th International Conference on Cutting-Edge Organic Chemistry in Asia Lectureship Award (2006); Japan/UK GSC Symposium Lectureship (2007); and the IBM Japan Science Prize (2007). His research interests include asymmetric catalysis, biomimetic catalysis induced by artificial enzymes, dehydrative condensation catalysis toward green and sustainable chemistry, and acid–base combination chemistry. ■



Scheme 1. Double activation of organometallic reagents and carbonyl compounds.

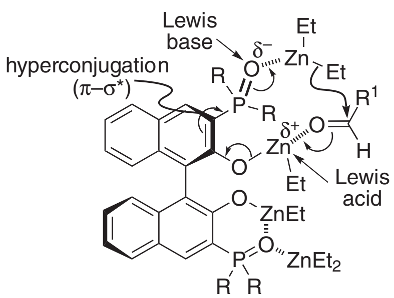
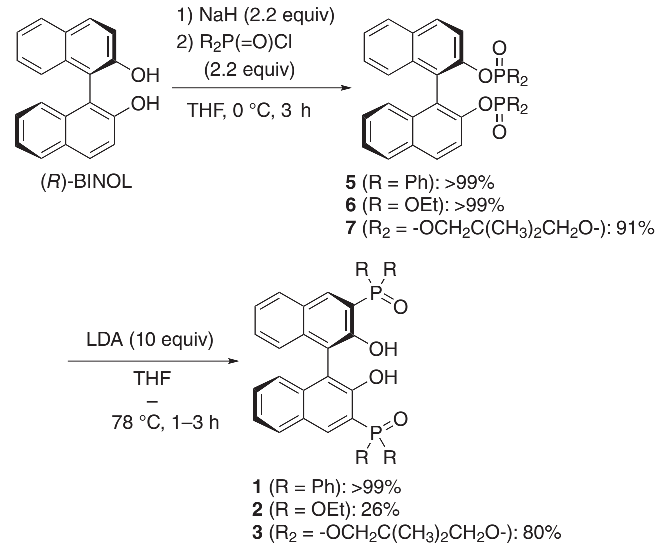


Fig. 1. Chiral 3,3′-diphosphoryl-BINOLate–Zn(II) complex as a conjugate Lewis acid–Lewis base catalyst.

BINOLate–Zn(II) catalysts for a highly enantioselective organozinc addition to aldehydes without $Ti(Oi\text{-}Pr)_4$. The key in designing our catalysts is conjugation between a 2-naphthol moiety and a 3-phosphoryl group (P═O) to doubly activate the electrophile and nucleophile.[17,18] In particular, we examined enantioselective organozinc addition to aldehydes using chiral BINOLate–Zn(II) catalysts bearing phosphine oxides [P(═O)R_2], phosphonates [P(═O)$(OR)_2$], or phosphoramides [P(═O)$(NR_2)_2$] at the 3,3′-positions (Fig. 1).

Preparation of (*R*)-3,3′-Diphosphoryl-BINOLs

(*R*)-3,3′-*bis*(diphenylphosphinoyl)-BINOL (**1**) was prepared almost quantitatively from commercially available (*R*)-BINOL



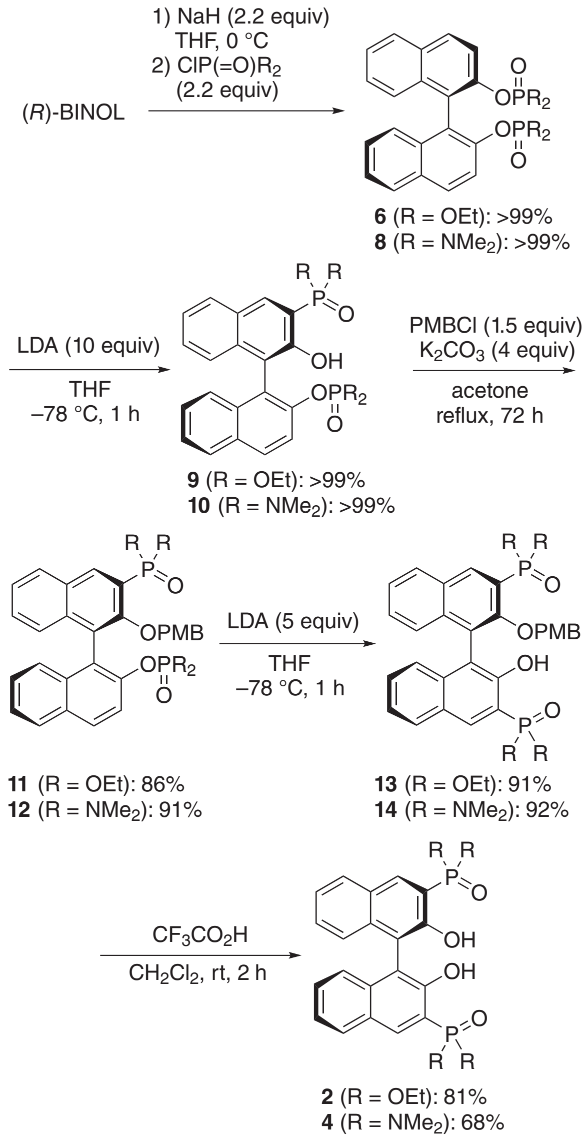
Scheme 2. Preparation of (*R*)-3,3′-diphosphoryl-BINOLs.

in two steps via phospho-Fries rearrangement (Scheme 2).[14a] The starting (*R*)-BINOL in THF was treated with NaH. Subsequent dropwise addition of diphenylphosphinic chloride gave the corresponding phosphinates **5** quantitatively without further purification. The rearrangement of **5** proceeded with LDA in THF at −78° C for 1 h to give **1** quantitatively as colorless crystals after recrystallization from toluene/hexane. Phosphonate [P(═O)$(OR)_2$] ligands **2** and **3**, which had substituents that were more electron donating than the phenyl groups in **1**, were also prepared in a similar manner via phospho-Fries rearrangement of **6** and **7**.

Unfortunately, our promising **4** could not be prepared by the simple procedures described earlier because the key phospho-Fries rearrangement could not proceed on both sides of the 3- and 3′-positions. However, protection of 2-naphthol in **10** with a PMB (*p*-methoxybenzyl) group after the first phospho-Fries rearrangement enabled the second rearrangement to **14**, and **4** was obtained after deprotection of the PMB group (Scheme 3).[14b,c] This synthetic method was also effective for preparing **2**, which was obtained in low yield by a usual phospho-Fries rearrangement, as shown in Scheme 2. The total yields of **2** were improved to 59% in Scheme 3.

Evaluation of Conjugate Acid–Base Catalyst, 3,3′-Diphosphoryl-BINOLate–Zn(II) Complexes

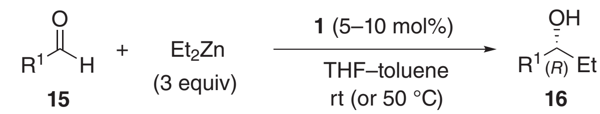
First, 10 mol % of **1** was used to catalyze the addition of diethylzinc (3 equiv) to aldehydes (**15**) in THF–toluene (1 : 1)



Scheme 3. Preparation of (*R*)-3,3′-diphosphoryl-BINOLs.

at room temperature. The results are summarized in Table 1.[14a] In particular, aromatic aldehydes with electron-donating or electron-withdrawing groups showed high enantioselectivities (up to 95% ee) (entries 1–4). The reactions also proceeded smoothly for heteroaromatic aldehydes and α,β-unsaturated aldehydes (entries 5–8). For aliphatic aldehydes, in which competitive reduction often occurs along with ethylation, the corresponding secondary aliphatic alcohols were obtained with high enantioselectivities (up to 94% ee) (entries 9–11). An increase in the catalytic activity of **1** was observed at 50° C with a slight loss of enantioselectivity, and the catalyst loading could be decreased to 5 mol % (parentheses in Table 1).

Table 1. Catalytic enantioselective diethylzinc addition to aldehydes using **1**.



Entry	R^1	Time (h)[a]	Yield (%)[a]	ee (%)[a]
1	Ph	3 (2)	95 (84)	95 (88)
2	4-$MeOC_6H_4$	18 (7)	89 (95)	89 (85)
3	2-FC_6H_4	9	89	86
4	4-ClC_6H_4	1 (0.5)	98 (92)	94 (90)
5	2-Furyl	3 (1.5)	90 (93)	84 (80)
6	3-Thienyl	2 (1.5)	94 (95)	92 (83)
7	PhC≡C	1	86	86
8	(E)-PhCH═CH	24	82	75
9	$PhCH_2CH_2$	6 (4)	73 (50)	82 (86)
10	*n*-C_5H_{11}	12	71	94
11	*n*-C_9H_{19}	12	72	90

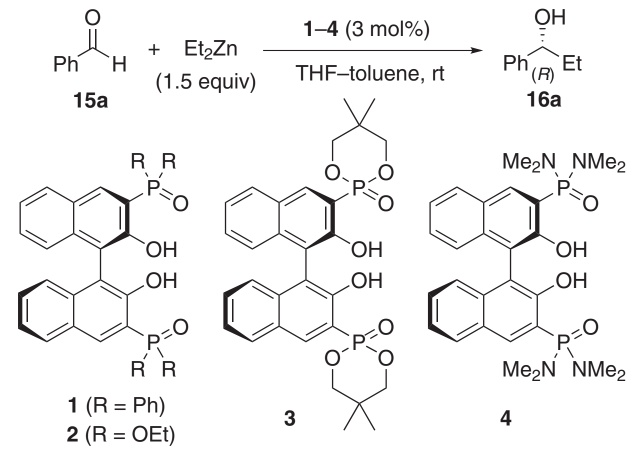
[a]5 mol % of **1** and 3 equiv of Et_2Zn were used at 50° C (data in parentheses).

BINOLate–Zn(II) Catalysts Bearing Phosphonates or Phosphoramides

To achieve a higher Lewis basicity of phosphoryl moiety to increase the nucleophilicity of organozinc reagents, chiral ligands (**2** and **3**), which had electron-donating *P*-substitutions, were examined (Table 2).[14b,c] The reaction with 3 mol % of **1** proceeded with high enantioselectivity (93% ee), but was slow. However, even 3 mol % of **2** or **3** was more effective than **1** for the enantioselective addition of diethylzinc (1.5 equiv) to benzaldehyde (**15a**) in THF–toluene at room temperature (entries 1–3). This preference for **3** to **2** was probably because of the strong electron donation and compact structure required by a fixed ring conformation. For a promising ligand with a higher Lewis basicity, we next examined a BINOLate–Zn(II) catalyst bearing conjugate phosphoramide moieties [P═(O)(NMe_2)$_2$]. The reaction proceeded smoothly with 3 mol % of **4** and 1.5 equiv of Et_2Zn at room temperature for 24 h to give the product in 98% yield with 97% ee (entry 4). Therefore, the catalytic activity of **4** was extremely higher than those of **1**, **2**, and **3**, and the order of reactivity was **4** > **3** > **2** > **1**, according to their electron-donating abilities to phosphoryl moieties.[19] Using **1**–**4** at a warm temperature (50° C) increased the reactivity with almost the same enantioselectivity (within 0–2% ee) (entries 5–8). In particular, by using **4**, the Et-adduct was obtained in quantitative yield with 95% ee within only 4 h (entry 8).

Encouraged by the preliminary performance of **4**–Zn(II) catalyst in the enantioselective ethylation of **15a**, we next

Table 2. Ligand optimization in catalytic enantioselective diethylzinc addition to benzaldehyde.



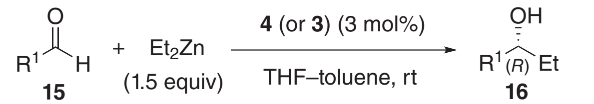
Entry	Ligand	Temperature	Time (h)	Yield (%)	ee (%)
1	**1**	rt	72	77	93
2	**2**	rt	48	70	94
3	**3**	rt	48	81	97
4	**4**	rt	24	98	97
5	**1**	50° C	12	60	91
6	**2**	50° C	12	65	94
7	**3**	50° C	12	76	96
8	**4**	50° C	4	>99	95

examined the generality of this method for other aldehydes (Table 3).[14b,c] For aromatic aldehydes, high enantioselectivities and high yields were observed under the optimized reaction conditions with 3 mol % of **4** and 1.5 equiv of Et_2Zn (entries 1–11). The general reactivities were higher with electron-withdrawing aromatic aldehydes than electron-donating ones, although high enantioselectivities were not affected by the type of substitution or its position. Heteroaromatic aldehyde, α,β-unsaturated aldehyde, and aliphatic aldehydes were also successfully applicable (entries 12–14). It should be noted that **3** was also useful because it showed high enantioselectivities and was easily prepared in two steps, in contrast to the more elaborate preparation of **4** (parentheses in Table 3).

Enantioselective *n*-Bu_2Zn and Me_2Zn Addition to Aldehydes

Other catalytic enantioselective additions of dialkylzinc to aldehydes were examined (Table 4).[14b,c] Despite the general difficulties in *n*-Bu-addition or Me-addition reactions, **4** (3–5 mol %) and 1.5–2 equiv of *n*-Bu_2Zn or Me_2Zn gave smooth conversion at room temperature. High enantioselectivities

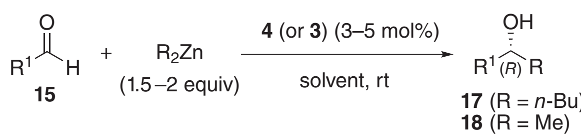
Table 3. Catalytic enantioselective diethylzinc addition to aldehydes using **4**.



Entry	R^1	Time (h)[a]	Yield (%)[a]	ee (%)[a]
1	Ph	24 (48)	98 (81)	97 (97)
2	4-$MeOC_6H_4$	48	91	94
3	2-MeC_6H_4	24	90	96
4	4-MeC_6H_4	24	90	96
5	4-PhC_6H_4	24	85	93
6	2-ClC_6H_4	12 (24)	92 (92)	94 (91)
7	3-ClC_6H_4	8 (24)	95 (96)	98 (96)
8	4-ClC_6H_4	12 (36)	93 (98)	95 (93)
9	4-$CF_3C_6H_4$	1 (3)	95 (95)	97 (97)
10	2-Naph	12 (24)	94 (87)	>99 (97)
11	3,4-$(OCH_2O)C_6H_3$	24 (48)	87 (86)	95 (90)
12	$PhC{\equiv}C$	12 (24)	70 (98)	79 (90)
13	*n*-C_5H_{11}	24	85	77
14	3-Thienyl	12 (48)	94 (86)	95 (95)

[a]3 mol % of **3** was used in place of **4** (data in parentheses).

Table 4. Catalytic enantioselective *n*-Bu_2Zn or Me_2Zn addition to aldehydes using **4**.



Entry	R^1	R	Time (h)[a]	Yield (%)[a]	ee (%)[a]
1	Ph	*n*-Bu	24 (48)	94 (92)	96 (93)
2	4-MeC_6H_4	*n*-Bu	72	78	95
3	3-ClC_6H_4	*n*-Bu	48	87	92
4	4-ClC_6H_4	*n*-Bu	24 (48)	94 (89)	90 (95)
5	4-$CF_3C_6H_4$	*n*-Bu	12 (24)	91 (98)	97 (93)
6[b]	4-$CF_3C_6H_4$	*n*-Bu	48	92	92
7	3-Thienyl	*n*-Bu	36	83	89
8	Ph	Me	72	82	96
9	4-MeC_6H_4	Me	72	80	90
10	3-ClC_6H_4	Me	72	84	87
11	4-ClC_6H_4	Me	48	90	95
12	4-$CF_3C_6H_4$	Me	24 (48)	88 (60)	91 (82)
13[c]	4-$CF_3C_6H_4$	Me	48	84	86

[a]5 mol % of **3** was used instead of **4** (data in parentheses). Unless otherwise noted, 5 mol % of **4** and 2 equiv of R_2Zn were used in THF–toluene for methylation and in THF–heptane for butylation.
[b]3 mol % of **4** and 1.5 equiv of *n*-Bu_2Zn were used.
[c]3 mol % of **4** and 2 equiv of Me_2Zn were used.

were observed in the aromatic aldehydes without competitive reductions into $ArCH_2OH$. In particular, dialkylzinc addition to **15a** with 5 mol % of **4** showed high enantioselectivities (96% ee, entries 1 and 8). In place of **4**, 5 mol % of **3** was also effective for both *n*-Bu-addition or Me-addition to aldehydes, and the corresponding products were obtained with high enantioselectivities (82–95% ee), although longer reaction times were necessary (parentheses in Table 4).

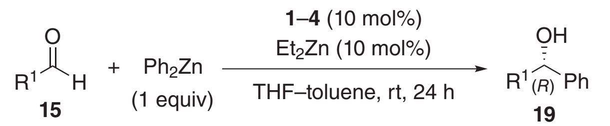
Enantioselective Ph_2Zn Addition to Aldehydes

Catalytic enantioselective diphenylzinc addition to **15** was examined in the presence of conjugate Lewis acid–Lewis base BINOLate–Zn(II) catalyst and 10 mol % of Et_2Zn.[14a,14c,20] Unexpectedly, low yield and low enantioselectivity were observed in the phenylation of 4′-chlorobenzaldehyde when **4**, **3**, and **2** were used (Table 5, entries 1–3). However, **1** showed good reactivities, and the corresponding Ph-adduct was obtained in 96% yield and 88% ee (entry 4). For other aromatic aldehydes bearing electron-donating or electron-withdrawing groups, **1** was so effective that the reactions proceeded smoothly to give the corresponding 1,1-diarylcarbinols with 73–88% ee. The reason for the unexpected reversal of catalytic activity between phenylations (**1** > **2** > **3** > **4**) and alkylations (**4** > **3** > **2** > **1**) is not clear, and further examinations are necessary. For one reason, a steric factor involving π–π interactions among the catalysts and Ph_2Zn cannot be denied. Eventually, our Lewis acid–Lewis base BINOLate–Zn(II) catalysts [i.e., **1**–Zn(II) and **4**–Zn(II)] gave complementary results in enantioselective phenylation and alkylation, respectively.

Characteristics of Zn(II) Cluster and Dinuclear Zn(II) Complexes Through Stoichiometric Reaction

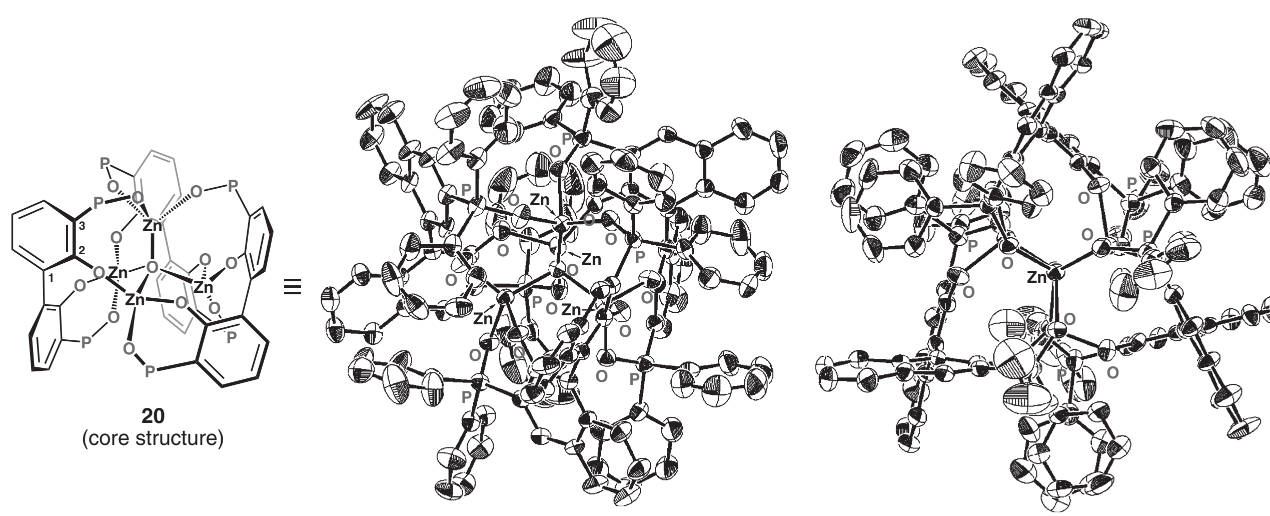
The key to characterizing the Zn(II) catalysts should be to clarify whether the phosphoryl moiety coordinates to the Zn(II) center as a Lewis base. Fortunately, a single crystal was obtained for X-ray analysis from a mixture of **1** and Et_2Zn (1 equiv each) in dichloromethane–hexane at room temperature for 24 h under *open-air conditions*.[14a] The ORTEP drawings are shown in Figure 2. The structure of the obtained Zn(II)

Table 5. Catalytic enantioselective diphenylzinc addition to aldehydes using **1**.



Entry	Ligand	R^1	Yield (%)	ee (%)
1[a]	4	4-ClC_6H_4	65	49
2[a]	3	4-ClC_6H_4	62	52
3	2	4-ClC_6H_4	78	87
4	1	4-ClC_6H_4	96	88
5	1	4-FC_6H_4	>99	86
6	1	$4\text{-CF}_3C_6H_4$	93	88
7	1	4-MeC_6H_4	86	85
8	1	4-PhC_6H_4	93	82
9	1	4-MeOC_6H_4	77	73
10	1	1-Naph	97	77
11	1	2-Naph	95	81

[a]Reaction time was 48 h.

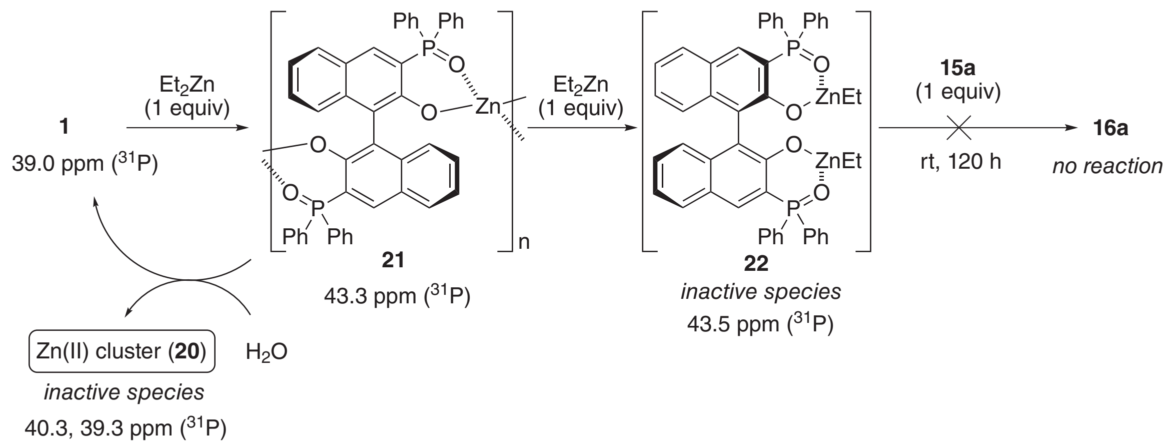


20
(core structure)

Side view

Top view

Fig. 2. ORTEP drawings of Zn(II) cluster (**20**).



Scheme 4. Postulated oligonuclear and dinuclear Zn(II) complexes.

cluster **20** was self-assembled from four Zn(II) metals and three units of **1**. At the center of this cluster, μ_4-O, which might be derived from adventitious water during the recrystallization, coordinates to four Zn(II) centers. The Zn(II) center in the top position coordinates to μ_4-O and three P═O in three different units of **1**. Each of the other three Zn(II) centers at the bottom positions coordinates to μ_4-O, Naph—O, and O═P—C═C—O through chelation. The position of P═O was not in extension of the naphthyl plane, and the torsion angles of O═P—C(3)—C(2) were 50.9–58.1°. Interestingly, the R^*O_2Zn [$R^*(OH)_2$ = **1**] chelation to the Zn(II) center did not exist in **20**.

As expected, Zn(II) cluster **20** was not an active catalyst; ethylation of **15a** with 3.3 mol % of isolated crystal of **20** under the typical reaction conditions at room temperature for 24 h gave **16a** in 38% yield with 20% ee. In fact, in a ^{31}P NMR study in $CDCl_3$ under *anhydrous conditions*, the addition of 1 equiv of diethylzinc to **1** led to a new complex **21** with a singlet peak at 43.3 ppm (major, >98%), with downfield shifts from 39.0 ppm for **1** (Scheme 4). Meanwhile, 2 equiv of ethane gas were trapped, and **20** was obtained as a minor product at 39.3 and 40.3 ppm (<2%). Complex **21** was highly moisture sensitive, and decomposed to the inactive complex **20** almost quantitatively with the partial release of **1** for 24 h under *open-air conditions*. Complex **21** with one singlet peak with large downfield shifts in the NMR study suggested a symmetrical structure with two phosphoryl units coordinating to Zn(II) centers, namely oligomeric [—OR*O—Zn—]$_n$ ($n \geq 2$) with Zn···O═P coordination but without R^*O_2Zn chelation. This postulated structure of **21** was also supported by X-ray analysis of **20**. Furthermore, the addition of another 1 equiv of Et_2Zn to **21** caused no release of gas to provide a probable C_2-symmetric dinuclear Zn(II) complex **22** [$R^*(OZnEt)_2$] with two independent NaphO–ZnEt···O═P chelations. The optical rotation ($[\alpha]_D^{21.0}$ in toluene) of **22**

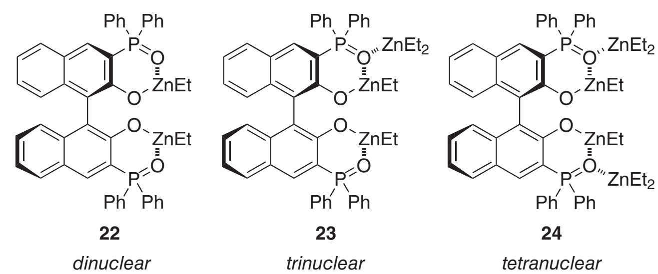


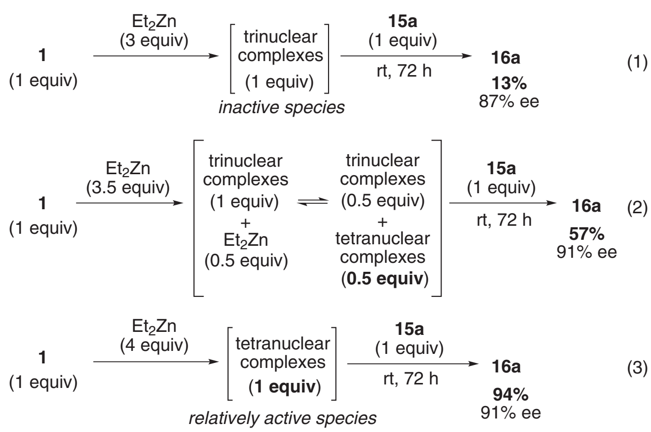
Fig. 3. Examples of possible Zn(II) complexes.

(−39.5°, c = 3.22) dramatically changed from that of **21** (−94.6°, c = 2.61), although **22** showed one major singlet peak at 43.5 ppm with almost no changes from **21**. The addition of **15a** (1 equiv) to dinuclear **22** (1 equiv), which was prepared by adding either directly 2 or 1 then 1 equiv of Et_2Zn to **1**, showed no reactivity, and therefore **22** was the inactive species.

Characteristics of Trinuclear and Tetranuclear Zn(II) Complexes Through Stoichiometric Reactions

The relationship between the ee values of **1** and **16a** was linear.[14a] This linear relationship suggests that, for one possibility, trinuclear monomeric species such as **23** by way of dinuclear **22** might be involved in the enantioselective addition of diethylzinc (Fig. 3). Unexpectedly, however, these assumed trinuclear complexes, which were directly prepared from **1** (1 equiv) and diethylzinc (3 equiv) with a major singlet peak at 38.0 ppm in ^{31}P NMR, showed low reactivity; the stoichiometric reaction to **15a** (1 equiv) gave **16a** in 13% yield with 87% ee after 72 h at room temperature (Scheme 5, Eq. 1).

However, the stoichiometric reaction with **15a** (1 equiv) and the complexes prepared from **1** (1 equiv) and diethylzinc (3.5 equiv) improved the yield of 57% (Eq. 2). From these results, trinuclear complexes would not be the active species, and the equilibrium between the trinuclear complex and another complex such as a tetranuclear one should be considered. In fact, the stoichiometric reaction with **15a** (1 equiv) and the expected tetranuclear complexes (1 equiv), which were prepared from **1** (1 equiv) and diethylzinc (4 equiv), could proceed and **16a** was obtained in 94% yield with 91% ee (Eq. 3). Interestingly, the reactivity under the stoichiometric conditions in Equation 3 was significantly lower than that under the catalytic conditions (vs. Table 1, entry 1). Thus, we can ultimately assume that the active species under the normal catalytic reaction conditions may be derived from the tetranuclear complexes such as **24** or likely the C_2-symmetric pentanuclear complex **25** (Scheme 6). In two postulated catalytic cycles: (i) inactive dinuclear **22** or trinuclear **23** would be regenerated via relatively active tetranuclear **24**, and (ii) relatively active tetranuclear **24** would be regenerated through much more active pentanuclear **25** after carbon–carbon bond formation with an aldehyde.[14c]



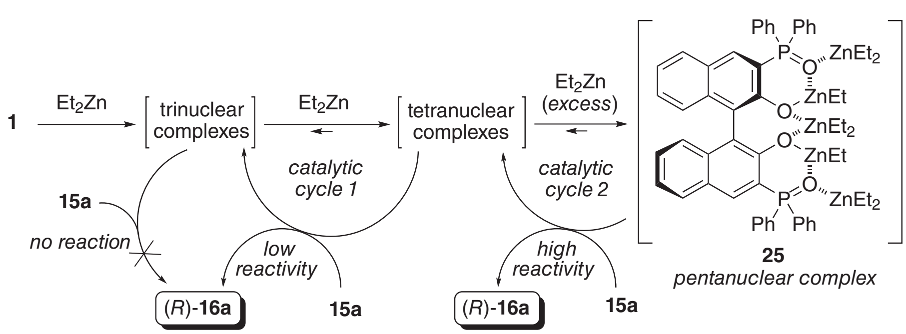
Scheme 5. Stoichiometric enantioselective diethylzinc addition.

Effect of Phenol and Biphenol Structures

Next, the effect of phenol and biphenol on the catalytic activity was examined.[14c] Catalytic diethylzinc addition to **15a** was explored with achiral ligands **26** and **27** (Scheme 6). The catalytic activity of **26** (20 mol %) with a phenol backbone was low, and 33% of **16a** was given for 24 h (Eq. 4). The low reactivity of **26** could probably be rationalized by a six-membered Zn(II) chelation as a stabilized structure. In high contrast, 10 mol % of **27** with a biphenol backbone could catalyze the reaction effectively, and **16a** was obtained in 89% for 6 h (Eq. 5). Therefore, the biphenol moiety might be an essential structure to enhance the reactivity. These results indicated that a new seven-membered chelation of diethylzinc with $R(OZnEt)_2$ [$R(OH)_2$ = biphenol] could trigger the dissociation of the original six-membered Zn(II) chelation to relieve steric hindrance before the transition states (Eq. 5).

Conformation of P═O and NaphO–Zn(II) Moieties

The direction of the phosphoryl moiety (P═O) in ligands **1**–**4** depends on the binaphthyl plane (Fig. 4): either (i) a flat conformation that has a π-orbital network and a six-membered ring by Zn(II) chelation, or (ii) a nonflat conformation with hyperconjugation through π(2-naphthol)–σ*(P—O) is possible. Therefore, a BINOL moiety is suitable for controlling the direction of conjugate P═O (Lewis base), which can be away from the Zn(II) ion (Lewis acid). To assist the proposed conformations such as shown in Figure 4A, an X-ray analysis of (*R*)-**1** is shown in Figure 5A.[14a,14c] Hydrogen bonding, which was observed in Naph—O—H···O═P (1.86 and 1.93 Å) to form a six-membered ring, supports the notion that the



Scheme 6. Proposed catalytic cycles.

chelation to Zn(II) (instead of H) is done in this manner. The hydrogen bonding in (*R*)-**1** was responsible for the observed large downfield shift, such as 10.57 ppm in ^{1}H NMR in $CDCl_3$. Interestingly, the positions of P═O were not in extension of the naphthyl plane, probably due to π(2-naphthol)–σ*(P—O) interaction; the torsion angles were 45.4° for C(2)—C(3)—P(1)—O(2) and 45.0° for C(12)—C(13)—P(2)—O(4). Fortunately, another conformation as shown in Figure 4B could be observed by X-ray analysis of (*S*)-**1** (Fig. 5B).[14c] In that crystal structure, hydrogen bonding was not observed in Naph—O—H⋯O═P. The torsion angles were 67.9° for C(2)—C(3)—P(1)—O(2) and 66.6° for C(12)—C(13)—P(2)—O(4), which were larger than were those of (*R*)-**1** involving hydrogen bondings. These structures of (*R*)-**1** with hydrogen bondings and (*S*)-**1** without hydrogen bondings might support the existence of π(2-naphthol)–σ*(P—O) interaction.

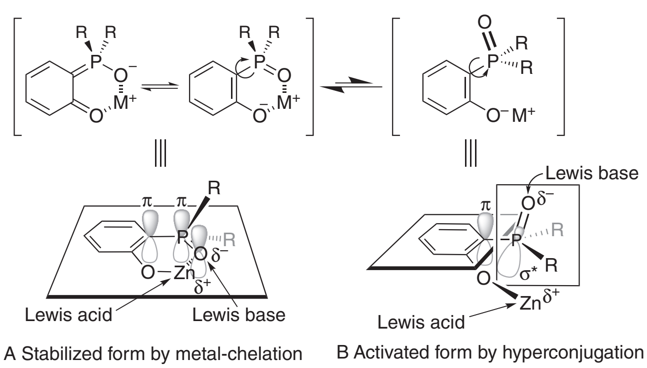


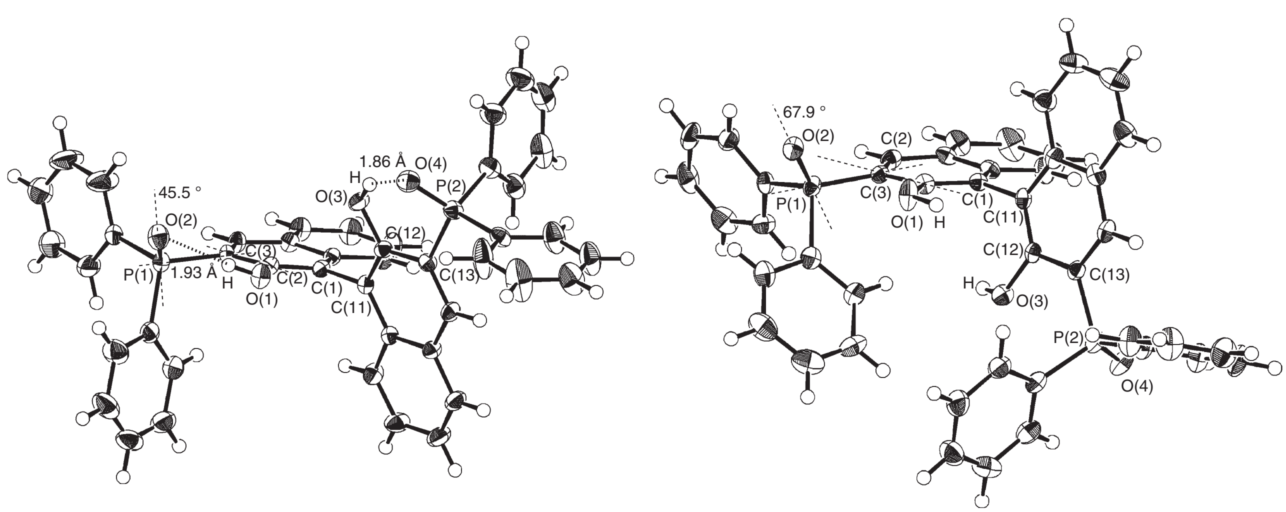
Fig. 4. Conjugate acid–base catalyst with a phosphoryl moiety in a BINOL backbone.

Proposed Transition State Assembly With Conjugate Acid–Base BINOLate–Zn(II) Catalysts

The mechanistic aspect of the transition state assembly, which should be regarded as a working model, is considered (Fig. 6).[14c] Based on the previous examinations, the active Zn(II) complex in this catalysis is likely to be monomeric tetra- or pentanuclear species (Scheme 6). The proposed transition state assembly (**28**) may provide the conformation as shown in Figure 4B with π(2-naphthol) orbital interactions to σ*(P—O) through hyperconjugation. Compound **28** has two conjugate Lewis acidic NaphO–Zn(II)–Et centers to activate aldehyde and Lewis basic phosphoryl groups to activate diethylzinc. From the results of the catalysis by phenol and biphenol (Scheme 7), tetranuclear complexes might change to pentanuclear complex **25**, and then a dissociation of P═O⋯Zn in **25** would lead to **28**. Due to the steric hindrance between aldehyde and two substituents to phosphoryl moiety, *re*-face attack should be favored exclusively. Eventually, (*R*)-products can be obtained along with the regeneration of the tetra- and pentanuclear species.

Design of Conjugate Phosphoramide–Zn(II) Catalysts for Tertiary Alcohol Synthesis[21]

In the previous preliminary study of catalytic enantioselective organozinc addition to aldehydes leading to secondary



A (*R*)-**1** with hydrogen bondings B (*S*)-**1** without hydrogen bondings

Fig. 5. ORTEP drawing of (*R*)-**1** and (*S*)-**1**.

alcohols, a chiral BINOLate–Zn(II) bearing phosphoramides at the 3,3′-positions was very effective as a conjugate Lewis acid–Lewis base catalyst. Unfortunately, however, this BINOLate–Zn(II) did not catalyze the reaction of ketones and organozinc reagents leading to tertiary alcohols, and a significant amount of the aldol products was obtained instead of the desired products. In fact, there are few examples of catalytic enantioselective organozinc addition to ketones, in sharp contrast to aldehydes, due to steric and electronic constraints regarding the substrates and/or reagents. Hence, it is still a challenge to establish a high level of asymmetric induction as well as a high yield.[5–10] Thus, we devised highly active and simple chiral phosphoramide–Zn(II) complexes as conjugate Lewis acid–Lewis base catalysts for enantioselective organozinc addition to ketones to obtain the desired tertiary alcohols in high to excellent yields (Fig. 7).[22] These chiral Zn(II) catalysts, which are simply prepared *in situ*, are derived from an inexpensive natural amino acid (i.e., L-valine).

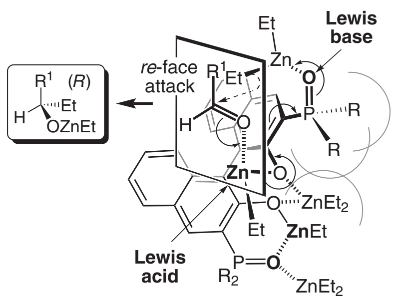


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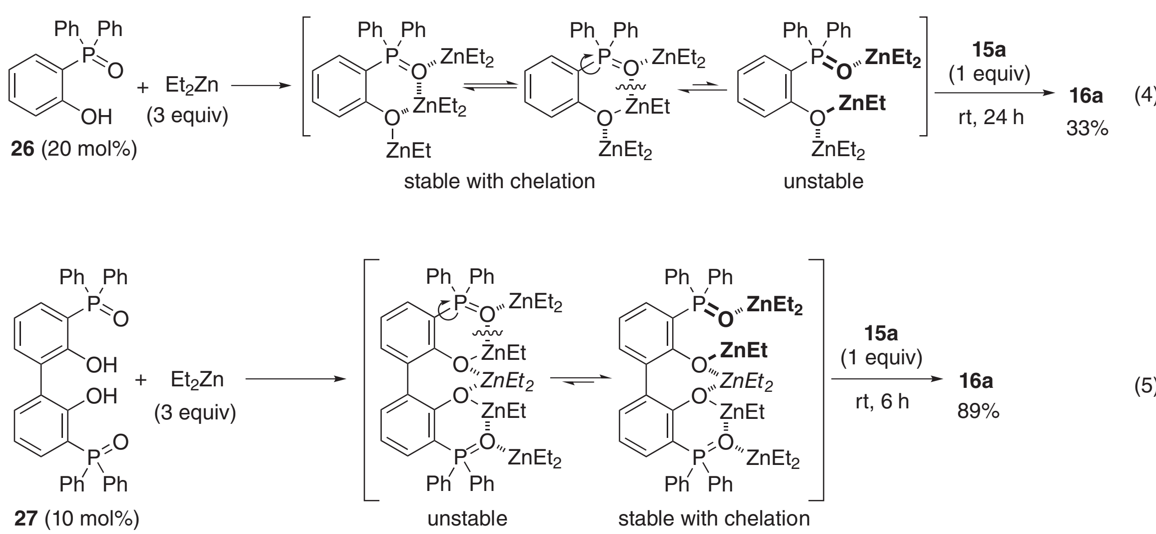
Fig. 6. Conjugate acid–base catalyst with a phosphoryl moiety in a BINOL backbone.

Evaluation of Conjugate Acid–Base Catalyst, Chiral Phosphoramide–Zn(II)

The chiral phosphoramide ligands (**30**–**33**) were readily prepared from commercially available *N*-Boc-protected L-valine (**29**) (Scheme 8). In each step, good to excellent yields were observed when diphenylphosphinic chloride [$Ph_2P(O)Cl$] was used. To develop conjugate Lewis acid–Lewis base catalysts suitable for tertiary alcohol synthesis from ketones, we initially investigated L-valine-derived chiral phosphoramide ligand (**30**, 10 mol %) in the catalytic enantioselective phenylation of 4′-chloroacetophenone (**34a**) with diphenylzinc (1 equiv) and diethylzinc (2 equiv) in heptane at room temperature (Table 6).[21] Fortunately, simply designed **30** gave **35a** with high enantioselectivity (80% ee), although the yield was low (30%). Chiral ligand **31** or **32** bearing a pyrrolidinyl or piperidinyl group instead of a NMe_2 group in **30** showed improved catalytic activity, but the enantioselectivity was the same or less. However, **33** bearing a sterically demanding *P*-(1-naphthyl) moiety showed considerable improvement in both yield and enantioselectivity (up to 95% ee). After optimization, **35a** was obtained in 98% yield with 92% ee by using 1 mol % of **33**.

We next examined the generality of this catalysis for other ketones (Table 7).[21] For aryl ketones with either an electron-withdrawing or electron-donating group, cyclic ketones, heteroaryl ketones, and α,β-unsaturated ketone with 10 mol % of (*S*)-**33**, high enantioselectivities (91–98% ee), and high yields (up to 98%) were observed in the products. Moreover, the aliphatic ketones also provided the corresponding tertiary alcohols with 80–82% ee in high yields. In particular, a 1 g scale (5 mmol) preparation of (*R*)-**35a** was established in 91%



Scheme 7. Effect of phenol and biphenol structures.

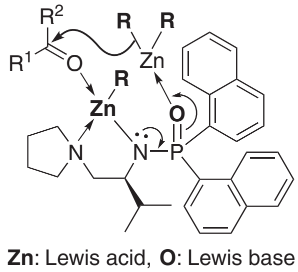
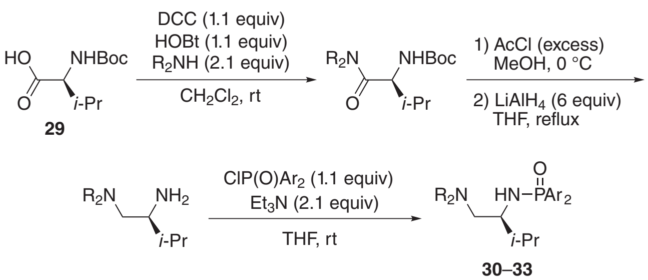


Fig. 7. Chiral phosphoramide–Zn(II) complex as a conjugate Lewis acid–Lewis base catalyst.



Scheme 8. Preparation of chiral phosphoramide ligand from L-valine.

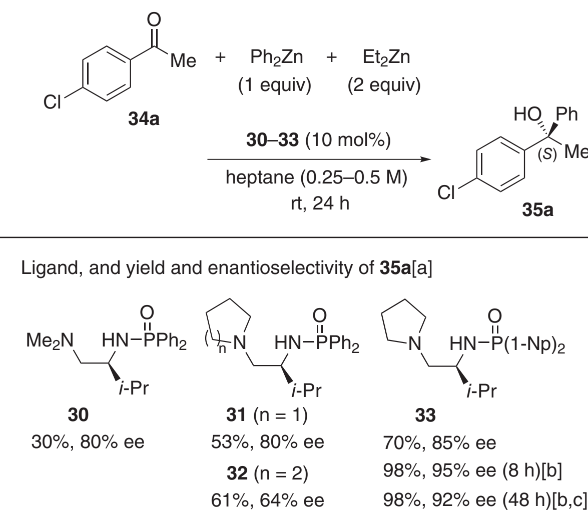
yield with 93% ee by using 3 mol % of (*R*)-**33** in heptane at room temperature for 24 h, which is the key intermediate in the synthesis of the antihistamine drug clemastine (**36**) (Scheme 9).[23]

Encouraged by the efficient phenylation to ketones, we next examined diethylzinc addition (Table 8).[21] As expected, the reaction of aryl- and heteroaryl methyl ketones with 3 equiv of diethylzinc proceeded smoothly at room temperature for 16–24 h in the presence of 10 mol % of **33**, and the desired tertiary alcohols were obtained in high yields with high enantioselectivities (91–96% ee). This is the first example of the highly efficient ethylation of ketones under mild reaction conditions by simply mixing a substrate, diethylzinc, and the chiral ligand in a solvent.

Proposed Transition State Assembly With Chiral Conjugate Phosphoramide–Zn(II)

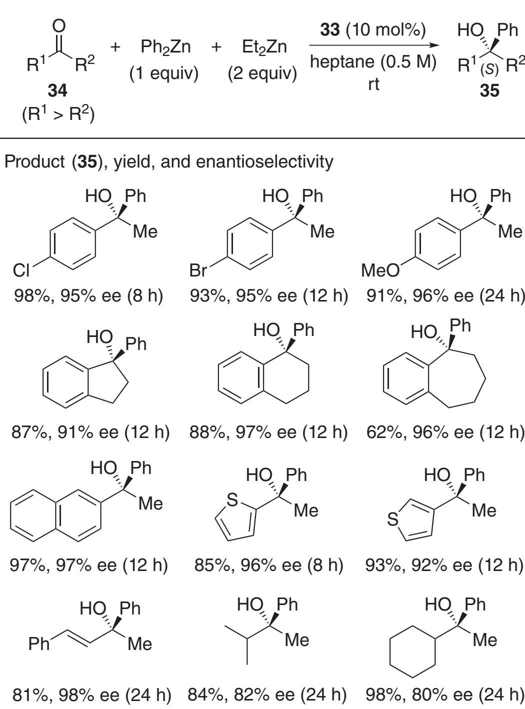
Finally, plausible transition state assemblies in catalytic enantioselective ethylations are shown in Figure 8, which should be regarded as a working model.[21] Chiral ligand (*S*)-**33** and diethylzinc would make up the *N*,*N*-chelated EtZn(II) center as a Lewis acid to activate a ketone (**34**, R^1COR^2, $R^1 > R^2$). The P═O moiety as a Lewis base, which is electron-donatively activated through conjugation among Zn—N—P═O bonds, would coordinate to the Et_2Zn(II) reagent, leading to six-

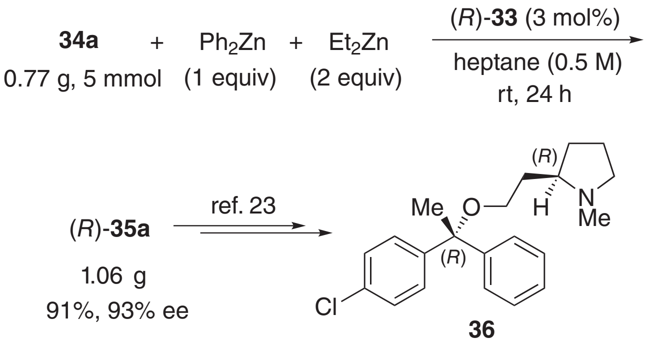
Table 6. Catalytic enantioselective phenylation of 4′-chloroacetophenone using phosphoramide–Zn(II) catalysts.



[a]Reactions were examined in 0.25 M heptane unless otherwise noted.
[b]Reactions were examined in 0.5 M heptane.
[c]1 mol % of **33** was used.

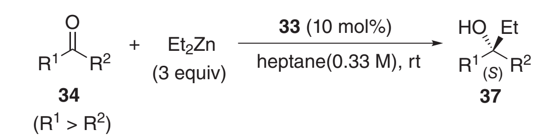
Table 7. Catalytic enantioselective phenylation of ketones using **33**.



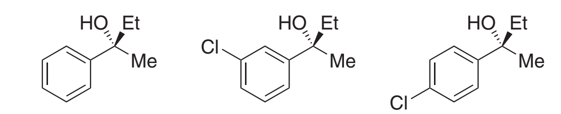


Scheme 9. Synthesis of clemastine intermediate.

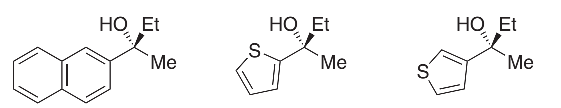
Table 8. Catalytic enantioselective diethylzinc addition to methyl ketones using **33**.



Product (**37**), yield, and enantioselectivity



80%, 93% ee (24 h) | 85%, 96% ee (16 h) | 77%, 92% ee (16 h)



76%, 96% ee (24 h) | 85%, 94% ee (24 h) | 77%, 91% ee (24 h)

membered chelation in a chair conformation with a ketone (**38** and **39**; see also Fig. 7). Cyclic amino groups of **31** and **32**, having more basic and less hindered characteristics due to electronic and steric factors, would smoothly coordinate to the Zn(II) center rather than the NMe_2 group of **30**. Thus, yields with **31** and **32** were improved from that with **30**, although a piperidinyl group of **32** might cause excessive hindrance to ketones leading to lower enantioselectivity (see Table 6). Unlike the *P*-phenyl moieties of **30**–**32**, sterically demanding *P*-(1-naphthyl) moieties of **33** would coordinate to the Zn(II) centers with a more appropriate conformation. Probably, the *i*-Pr moiety ("up" in Fig. 8) of the L-valine backbone might relay the direction of two 1-naphthyl moieties, "down" and "up," respectively. Ultimately, the bulkiness of "up"-1-naphthyl moiety would preferentially induce the coordination of the Et_2Zn reagent, which is activated by the P═O moiety, to the C═O moiety of **34**. Therefore, six-membered cyclic transition states would be stabilized by using **33**. In particular, the *si*-face

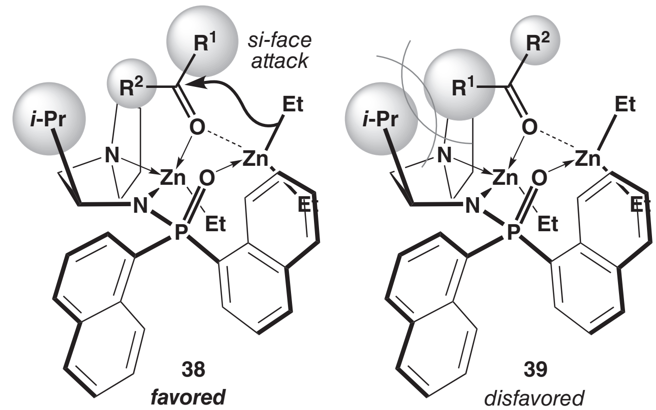


Fig. 8. Proposed transition states.

attack via **38**, which leads to (*S*)-product, should be favored exclusively without conspicuous steric repulsion between the *i*-Pr and pyrrolidinyl moieties of (*S*)-**33** and the R^1 group of **34**.

Conclusion

In summary, we have developed a highly efficient enantioselective organozinc addition to not only aldehydes but also ketones using conjugate Lewis acid–Lewis base catalysts. These chiral Zn(II) catalysts (i.e., chiral 3,3′-diphosphoryl-BINOLate–Zn(II) complex and chiral phosphoramide–Zn(II) complexes, which are simply prepared *in situ*) are derived from an inexpensive BINOL or L-valine. From a variety of aromatic and aliphatic ketones and aldehydes, optically active tertiary alcohols and secondary alcohols, respectively, were obtained in high yields with high enantioselectivities under mild reaction conditions without Ti(O*i*-Pr)$_4$. Further investigations to establish more efficient reaction conditions with minimum amounts of both organozinc reagents and catalysts are underway.

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